

Misfire and Engine Health Diagnostics

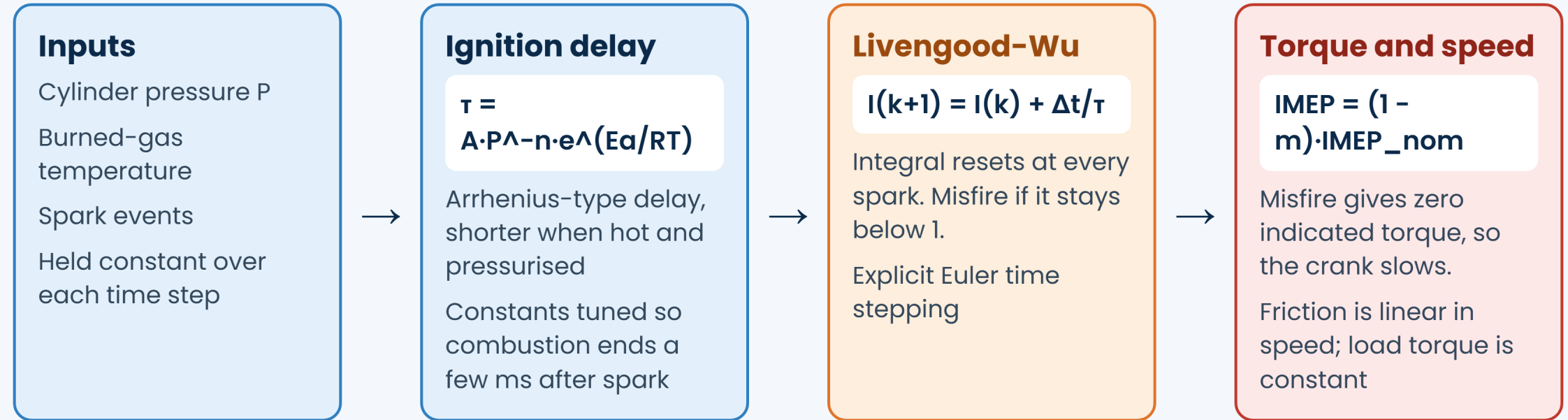
From physics-based models to early warning flags

Misfire model

Engine health model

Misfire Detection: Spark to Crank Speed

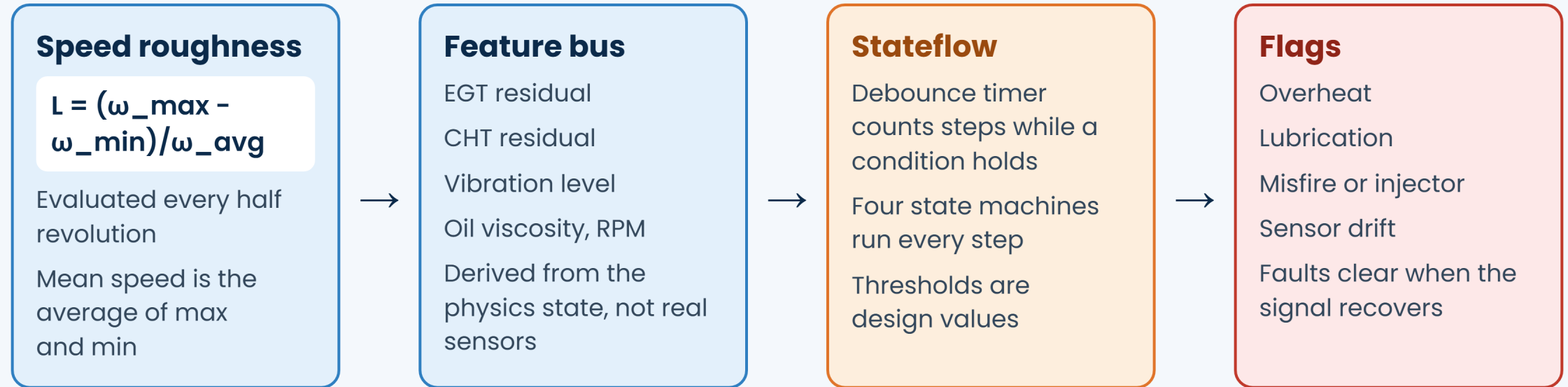
STAGES 1 AND 2 · IGNITION
AND TORQUE



No complete combustion means no torque, and the crank speed dips.

From Roughness to Diagnostic Flags

STAGES 3 TO 5 · SIGNALS AND STATEFLOW



Misfire flag: EGT residual and vibration both stay high for N steps.

Oil Health: Viscosity to Film Thickness

ENGINE HEALTH MODEL ·
STAGES 1 AND 2

Walther equation

$$\log \log(v + c) = A - B \cdot \log T$$

Pure-oil viscosity from sump temperature

Valid over the range, with a lower bound for hot oil



Fuel dilution

$$v = v_{\text{pure}} \cdot (1 - k \cdot f)$$

More fuel in oil means thinner oil

Viscosity loss is linear in dilution fraction f

Sommerfeld number

$$S = (r/c)^2 \cdot \mu N / P$$

Combines viscosity, shaft speed and bearing pressure

Pressure = load / projected area



Film thickness

$$h_0 = c \cdot (1 - \varepsilon)$$

Empirical eccentricity, then minimum film

Steady state, no shaft misalignment

Low film thickness or low S means boundary contact: a lubrication warning.

Knock, Sensor Drift and Failure Modes

ENGINE HEALTH MODEL ·
STAGES 3 TO 5

Knock / LSPI

$$\tau = \tau_0 \cdot (1 - k \cdot r)$$

Oil droplets shorten ignition delay
Livengood-Wu integral reaching 1 triggers knock
Fixed crank window; τ_0 has no pressure dependence

Sensor drift

MAP drift from oil heat
Lambda drift from oil ingress
EGT spike after knock
Drift changes readings, not the physical state

Failure modes

Full film, mixed or boundary
Knock: warning, then critical
Sensor and misfire faults
Thresholds are design limits

Oil ingress ties it together: bad lambda readings lead to fouling and misfire.

Physics-based residuals turn raw signals into early warnings.

Misfire model

Detects what combustion just did.

Engine health model

Tracks oil, knock and sensors.

Together

One flag set to act on before damage.